LALIT M

UNITY TECHNICAL LEAD | DIGITAL TWIN SOLUTION ARCHITECT

Chennai, India | +91 9042227145 | lalitsolanki0725@gmail.com | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Technical Lead & Senior Unity Developer bridging the gap between **Operational Technology (OT)** and high-fidelity 3D Visualisation. Architected a proprietary **Universal Field Bridge** for seamless **PLC data integration** (S7/Modbus/EtherNet/IP) and developed custom **Unity middleware** to handle high-frequency telemetry. Specialised in **performance profiling**, **custom editor tools**, and **modern UI architecture**, delivering highly responsive visualisations on resource-constrained hardware. Dedicated to engineering scalable, future-proof architectures that bridge the gap between physical operations and digital intelligence.

TECHNICAL SKILLS

- **Industrial Protocols:** OPC UA, Siemens S7 (S7Comm), EtherNet/IP, Modbus TCP/RTU, MQTT.
- **Core Development:** Unity 3D, C#, Python, Dart (Flutter), C++.
- **Architecture:** MVU, Dependency Injection (Zenject), Microservices, Configurable Data Pipelines.
- **Data & Networking:** SQLite (Edge Caching), MongoDB, TCP/IP, WebSockets, Socket.IO, Node.js, REST APIs.
- **Tools:** Git, Unity Cloud Build, Plastic SCM, Blender.
- **XR & UI:** AR Foundation, OpenXR, Immersal (Spatial Mapping), Unity UI Toolkit (USS/UXML).

PROFESSIONAL EXPERIENCE

Senior Unity Developer | Crion Technologies **July 2021 - Present**

- **OT Data Aggregation Framework:** Architected a configuration-driven middleware (Field Bridge) that unifies data from **Siemens S7, EtherNet/IP,** and **Modbus PLCs**. Enabled engineers to deploy **Mini-PC edge gateways** via JSON configs, buffering data in **SQLite** to ensure zero data loss before syncing to **MongoDB**.
- **High-Performance Unity Visualisation:** Engineered a high-performance **OPC UA visualisation pipeline**, utilizing **multi-threaded data processing** to offload heavy telemetry parsing from the main thread, ensuring smooth rendering of high-frequency data within the **Universal Render Pipeline (URP).**
- **Process Simulation Integration:** Integrated Process Simulators (DWSim) into Unity via external **DLLs** and **OPC UA** bridges, enabling real-time physics and chemical process validation within the **Digital Twin** environment.
- **Enterprise VR Training Ecosystem:** Architected a modular **VR Operator Training System** featuring a custom **Unity Editor Authoring Interface**. This **no/low-code** tool enables designers to visually construct complex step-by-step procedures without engineering support.
- **Offline Multiplayer:** Engineered a custom **TCP**-based networking layer specifically for secure, offline industrial environments. Built the entire stack to handle high-bandwidth **Quest-to-PC video streaming** and real-time state synchronization, allowing instructor monitoring without any internet or cloud dependency.

- **Real-Time Networking (Socket.IO):** Developed a resilient networking layer on top of **Socket.IO** to manage dynamic rooms and namespaces, optimizing data traffic to ensure robust connectivity and a **unified pipeline** for streaming high-frequency data.
- **Developer Experience (Editor Tools):** Developed a suite of productivity **micro-tools** (CSV-to-ScriptableObject converters, Bulk Tag Mapping) and many **Unity Editor Tools** that eliminated repetitive manual entry, significantly accelerating daily development velocity for the dev team.
- **AI R&D (Unity Sentis):** Experimented with leveraging Unity **Sentis** for **on-device inferencing**, enabling Digital Twin applications to run AI models locally without cloud dependencies.
- **Team and Technical Leadership:** Led technical mentorship across multiple teams and disciplines, guiding backend, frontend, and **Unity** developers on **SOLID design principles** and sustainable development practices. Transformed inconsistent workflows into a **scalable**, **standardised foundation**, improving overall quality through **hands-on guidance** and regular reviews.

Junior Unity Developer | TNQ Ingage — **Feb 2020 - May 2020**

- **VR Safety Simulation:** Developed immersive industrial safety training modules for **mobile VR** (Oculus/Daydream); optimised 3D assets and lighting to maintain stable frame rates on mobile hardware.
- **Localisation Systems:** Implemented a scalable **JSON**-based localisation system, enabling the seamless deployment of training apps across different regions.

AR Developer (Intern) | Flex India — **Jul 2019 - Sep 2019**

- **Remote Assistance AR:** Collaborated on an **AR remote support** tool for field technicians and developed the **UI** and interactive **Tutorial** systems.

EDUCATION

Bachelor of Science in Game Development — **2017 - 2020**

ICAT Design and Media College, Chennai

- **Specialization:** Gameplay Programming & Physics Simulation.

KEY PROJECTS

- **Architected** a runtime **Digital Twin** authoring tool enabling drag-and-drop creation of industrial assets and **live data binding** without rebuilds or developer involvement.
- Engineered a **geo-spatial** safety and zone-tracking engine to monitor personnel locations in real time, enforce **safe/danger** zones, and surface violations through **live dashboards.**
- Researched and prototyped a **Visual Positioning System (VPS)** leveraging Python and computer vision to process camera and GPS data, evaluating sparse **point cloud** generation for **indoor positioning** without physical markers.

ADDITIONAL INFORMATION

- **Languages:** English (Professional), Marwari (Native), Tamil (Fluent), Hindi (Fluent).
- **Hardware:** 3D Printing Enthusiast
- **IoT:** Industrial PLCs, Edge Gateways.
- **Utilities:** Building Productivity Micro-Apps.
- **Tools:** Postman (API Testing), Jira, Trello, Linear